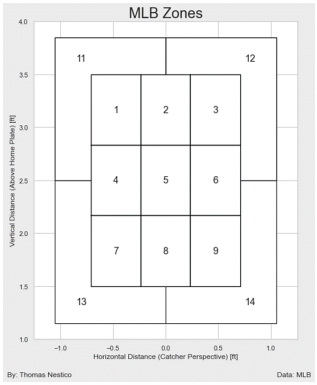
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Stats Final Project Codebook
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About the Data Source:
Baseball Savant uses Statcast (a high-speed, camera and radar tracking system installed in all 30 MLB ballparks) to obtain the pitch data. The technology tracks the pitches at 20,000 frames per second. The initial dataset was all pitches from the first week of the 2026 MLB Season (March 25th to April 1st). The analysis dataset only included pitches that were swung at and the following variables:

Variable Name	Variable Description	Type	Initial Values	Recoded Values	Source File
pitch_type	What type of pitch was thrown.	Cat	Ex. FF, SL, or CV (Fastball, Slider, or Curveball)	NA	Statcast
game_date	The date of the game.	Cat	Ex. 2026-04-01	NA	Statcast
player_name	The pitchers name.	Cat	Ex. Boyd, Matthew	NA	Statcast
batter	The batters 6 digit identification code.	Cat	Ex. 545361 (Mike Trout)	NA	Statcast
pitch_number	The number of pitches the pitcher had thrown so far in their outing	Numerical	Ex. 27	NA	Statcast
stand	The side of the plate the batter is standing on.	Cat	L or R	NA	Statcast
p_throws	The hand the pitcher throws with.	Cat	L or R	NA	Statcast
description	The description of the pitch outcome.	Cat	Ex. swinging_strike or hit_into_play	NA	Statcast
is_whiff	Whether the batter swung and missed, based on value in description.	Cat (Bin)	1=Swing and miss 0=Contact made with ball	NA	Added
effective_speed	The velocity of the pitch in miles per hour taking into account the pitcher's release extension. Calculated using basic kinematics. Corresponds to how	Numerical	Ex. 103	NA	Statcast

	much time the batter has to react to the pitch.				
zone	Where the pitch is located in relation to the batters handedness. Uses the standard zone depicted below:  But flips zones based on handedness of batter so zones 11, 1, 4, 7, and 13 are always inside.	Cat	1=Zone 1 2=Zone 2 3=Zone 3 (Depicted in image) etc.	For Right Handed Batter: 1=Zone 3 2=Zone 2 3=Zone 1 4=Zone 6 5=Zone 5 6=Zone 4 7=Zone 9 8=Zone 8 9=Zone 7 11= Zone 12 12=Zone 11 13=Zone 14 14=Zone 13 On image For Left Handed Batter: Same as image.	Statcast
pfx_x	The horizontal movement of the pitch in feet. It isolates spin-induced movement, comparing the actual pitch trajectory to a theoretical pitch thrown with no spin. Transformed so positive break is always away from the batter.	Numerical	Ex. 1.2	For Left Handed Batter: 1.2 = -1.2 For Right Handed Batter: 1.2 = 1.2	Statcast
pfx_z	The vertical movement of the pitch in feet. It isolates spin-induced movement, comparing the actual pitch trajectory to a theoretical pitch thrown with no spin.	Numerical	Ex. 1.4	NA	Statcast
plate_x	The horizontal position in feet of the baseball as it crosses the plate from the catcher's perspective. 0 is the center of the plate, negative values indicate the catchers left and positive values indicate the catchers right.	Numerical	Ex. 0.32	NA	Statcast

plate_z	The vertical position of the baseball as it crosses the plate. Measured in feet off of the ground.	Numerical	Ex. 2	NA	Statcast
same_hand	Whether the batter pitcher matchup is a same hand matchup. Derived from stand and p_throws.	Cat (bin)	1=Batter and Pitcher are both lefties or both righties 0=It is a righty lefty or lefty righty matchup	NA	Added
balls	The number of balls at the time of the pitch.	Cat	Ex. 3	NA	Statcast
strikes	The number of strikes at the time of the pitch.	Cat	Ex. 2	NA	Statcast
arm_angle	The angle of the pitcher's forearm relative to the ground at pitch release.	Numerical	Ex. 45	NA	Statcast
spin_axis	The observed orientation of the balls rotations at release	Numerical	Ex. 180	NA	Statcast
inferred_axis	The expected spin axis. Calculated using the pitches movement: arctan(-pfx_x/pfx_z)	Numerical	Ex. 135	NA	Added
spin_deviation	The difference between the spin axis and inferred_axis. Used to quantify other forces acting on the ball besides Magnus forces (forces from spin). The main other force is seam shifted wake which is a phenomenon caused by the ball's laces displacing air.	Numerical	Ex. 10 .5	NA	Added